Semax

(Intranasal: 150–900 mcg/day)

Semax is a synthetic ACTH-related heptapeptide intended to retain neuroactive effects without the steroidogenic activity observed with intact ACTH. [9,45] Semax has potential applications in neurological recovery, attention and memory, stress regulation, and selected optic-nerve disorders. Limited clinical studies and convergent animal experiments support likely adjunctive benefit for selected post-stroke functional outcomes; cognitive and ophthalmic applications require narrower, outcome-specific interpretation. [3–7,20,43,46,63] Regional and time-dependent neurotrophin, neurotransmitter, and injury-response effects provide a plausible basis for synaptic plasticity and neuroprotection. [11–18,24–31] Community descriptions of focus, mental clarity, and a calm cognitive effect are anecdotal.

💉 💉 Likely reasonably tolerated during short intranasal courses in selected patients; long-term safety unresolved. [5,8,51,60–62]

⭐ ⭐ Likely adjunctive benefit for selected post-stroke functional outcomes; possible benefit in selected optic-nerve disorders. [3–7,63]

⭐ Possible support for attention and short-term memory during demanding mental work, based on a small controlled operator study. [43]

⭐ Antistress effects demonstrated in several rodent models; corresponding human mood and resilience benefits are plausible. [32–36]

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

Short clinical courses were generally reported as well tolerated, including no reported adverse effects in the 110-patient rehabilitation study, but adverse-event ascertainment was limited. [5,8,62] A small mouse supporting experiment found no significant differences in selected liver and kidney markers three days after treatment in sham-operated animals. [39] Older experiments found no appreciable steroidogenic or melanocyte-stimulating activity under the tested conditions, supporting distinction from intact ACTH. [45]

Dose & Patterns

Animal distribution experiments and acute human functional-imaging changes suggest intranasal Semax can rapidly engage CNS-related pathways. [1,2,9] The licensed formulations described below are nasal drops; a spray cannot be assumed to deliver the same amount per actuation. [60,61]

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- **400–900 mcg/day intranasally for 3–5 days, with repeat courses described in the Russian prescribing information — mental-fatigue prevention regimen. [60]**
- **6 mg/day intranasally for 10 days, followed by a second 10-day course after a 20-day interval — studied during post-stroke rehabilitation. [5]**
- **12 mg/day intranasally for moderate stroke for five days, or 18 mg/day for severe stroke for ten days — reported in an acute ischemic-stroke study. [3]**

These higher-dose clinical regimens were adjuncts to condition-specific stroke care. Five months of follow-up in the rehabilitation study did not represent five months of continuous treatment. [3,5]

The speculative, experimental community cognitive-use intranasal patterns and doses discussed are

- **300–900 mcg/day in divided intranasal administrations**

Cycling at the community dose is speculative, intended to limit continuous exposure and preserve perceived benefits. Commonly discussed speculative patterns include:

- Occasional cognitive use: short, intermittent exposure
- Repeated community cycles: two to four weeks of use followed by four to eight weeks without use

Benefits

- **Ischemic-stroke recovery**: likely adjunctive benefit for selected functional outcomes, supported by limited human studies and convergent experimental neuroprotection. [3–5,20,46,47] In a 110-patient rehabilitation study, treatment was randomized within early- and late-rehabilitation groups; Semax was associated with faster improvement in Barthel daily-function scores. Timing imbalance in the late groups and incompletely described blinding limit causal precision. [5] Russian 2024 guidance includes Semax during rehabilitation stages 2 and 3, with recommendation grade B and evidence level 3. [63]
- **Spinal cord injury**: functional recovery improved in a female-mouse injury experiment, with accompanying cellular evidence involving lysosomal integrity and pyroptosis. [39] A rat experiment measured early cytokine changes rather than motor recovery. [50] Translation to human neurological recovery is plausible; the two studies do not constitute independent replication of functional benefit.
- **Alzheimer's disease models**: intranasal Semax improved selected memory outcomes and reduced amyloid plaque counts in one transgenic-mouse study. Behavioral effects were not uniform across tasks and testing days, and locomotor changes complicate interpretation. [37] Copper–amyloid experiments provide complementary cellular and physicochemical support for a plausible neuroprotective application. [38,49]

- **Learning and memory**: improvements have been observed in selected animal learning and post-ischemic memory tasks. [13,19,20,46] A double-blind experiment involving 16 male operators found better preservation of selected task performance after intranasal treatment. Its small, specialized population supports possible cognitive benefit during demanding work rather than a general enhancement claim. [43]
- **Neurotrophin signaling**: Semax alters BDNF/TrkB and NGF-related signaling in responsive neural tissues. The direction depends on region and timing, including transient decreases in some experiments. [11–15,21,22] Plasma BDNF increased in the rehabilitation study; its association with recovery does not establish that the biomarker change mediated the functional benefit. [5]
- **Mood and stress resilience**: multiple rodent experiments report attenuation of stress-related behavioral or neurochemical changes. [32–36] In the chronic-stress study, anhedonia and several stress-associated measures improved, but forced-swim immobility did not. [36] Human stress-regulation benefit is plausible, with translation limited by differences in model, endpoint, and exposure. Community mood benefits remain anecdotal.

Applications

- **Ischemic stroke** [3–5,63]
- **Short-term cognitive support** [43]
- **Optic-nerve disorders** [6,7,58,59,65]
- **Traumatic brain injury** [18,44,64]

Cautions

- **Pregnancy and breastfeeding**: contraindicated in the Russian prescribing information; reproductive and developmental safety are unresolved. [60–62]
- **History of seizures**: contraindicated in the Russian prescribing information. Paroxysmal EEG activity was reported in some patients in a posthypoxic-encephalopathy series, reinforcing this precaution. [51,60,61]
- **Acute psychiatric conditions and anxiety disorders**: listed contraindications. Known hypersensitivity is also a contraindication. [60,61]
- **Psychoactive medicines, including stimulants and monoamine oxidase inhibitors**: interactions affecting mood, sleep, or monoaminergic signaling are plausible. Semax potentiated amphetamine-associated dopamine and locomotor responses in rodents; this is not a demonstrated clinical interaction rate. [16] Anticoagulant or antiplatelet treatment warrants attention because experimental hemostatic effects make additive bleeding risk plausible. [41,62]

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Monitor For

- **Mood and behavior**: unusual agitation, elevated mood, impulsivity, or marked behavioral changes warrant reassessment; these are precautionary monitoring targets, not established incidence estimates. [16,51,60,61]
- **Sleep and neurological symptoms**: persistent insomnia, disruptive sleep changes, or seizure-like symptoms warrant attention. New bruising or bleeding is also relevant to the experimental hemostatic signal. [41,51,62]

Side effects

- **Sleep effects**: anecdotal community reports describe insomnia, disrupted sleep, or vivid dreams, particularly with late-day use; their frequency and attribution are uncertain.
- **Mood effects**: anecdotal community reports describe anxiety, agitation, or mood lability. These reports are distinct from the label contraindications for acute psychiatric conditions and anxiety disorders. [60,61]
- **Neurological and cardiovascular safety**: the seizure-related signal and psychiatric contraindications argue against a blanket continuous-use safety inference. Small clinical studies do not adequately resolve arrhythmia, blood-pressure, rare neurological, or cumulative neurobehavioral risks. [51,60–62]

Biological Mechanism

Semax is an ACTH-related peptide with rapid intranasal CNS-related activity and context-dependent neurotrophin signaling. Human imaging demonstrates network changes, while animal and cell experiments show changes in BDNF, TrkB, and NGF. [1,2,9,11–15] Neither uniform neurotrophin elevation nor a single established receptor mechanism captures these findings. Early reductions in hippocampal or retinal neurotrophin transcripts coexist with increases elsewhere or at other times. [14,15]

Monoamine and ion-channel effects are selective. Semax altered serotonin metabolism and potentiated amphetamine-associated dopamine responses in rodents, while Semax alone did not increase dopamine in the tested striatal preparation. [16] Laboratory studies report modest glutamate-receptor binding effects, delayed calcium or mitochondrial disruption during glutamate toxicity, enhanced GABA-activated currents in one neuronal preparation, and reduced glycine-activated currents in another. [17,18,48] Calcium oscillations and GABA-binding responses also depend on the experimental context. [40,42] These observations support plausible modulation of excitability rather than uniform transmitter stimulation.

After experimental cerebral ischemia, Semax changes neurotrophin, vascular, immune, and neurotransmission-related gene expression. Early immune-gene activation and later suppression of some inflammatory pathways are both reported; direction depends on model,

region, and sampling time. [21–31] Related transcriptomic reports can reuse animals and should not be counted as independent efficacy replications. [30,31] Effects on transcription also occur in uninjured rat brain. [66] PGC-1α-associated mitochondrial responses provide a further experimental neuroprotection mechanism. [47]

In spinal-cord injury models, the proposed pathway involves lysosomal membrane integrity, reduced pyroptosis, and USP18/FTO regulation. Computational μ-opioid receptor targeting is a mechanistic hypothesis rather than proof of direct binding or clinical opioid activity. [39] Copper binding, altered amyloid aggregation, and reduced copper-associated oxidative toxicity provide a separate cellular rationale for neuroprotection. [38,49] Together these pathways plausibly influence neuronal survival, plasticity, memory consolidation, stress adaptation, and recovery after injury.

Combining Selank and Semax

Concurrent or sequential Semax and Selank use is speculative community practice. Different but overlapping neurotrophin, monoamine, GABAergic, and immune effects make complementary cognitive or stress-regulation effects plausible. [2,11–16,24–31,42,48]

A comparative Parkinson-like rat experiment found no motor or passive-defensive benefit with either peptide, illustrating that mechanistic overlap does not guarantee functional benefit in every disease model. [57]

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